CES 2015 Data Set

No seed was used

"Harmony in Diversity: Unraveling the Tapestry of Same-Sex Marriage Perspectives in Canada"

Introduction

In 2017, tens of thousands of people protested against same-sex marriage in Australia, prior to the country's successful legalization of same-sex marriage. These protests illustrate the divisive nature of same-sex marriage as a social and political issue, and the strong emotions and beliefs it can evoke among people. The extent to which these protesting citizens are educated is a phenomenon that should be further explored. The main research question that this data analysis looks to explore is: Are citizens' attitudes toward same-sex marriage influenced by their level of education? This paper also explores the impact of other potentially relevant variables such as political party vote leaning, gender, and the concept of trust.

Education appears to be associated with support for same-sex marriage. Same-sex marriage is more likely to be supported by people with a higher level of education. There is a correlation between higher education and critical thinking skills, as well as a progressive ideology that supports same-sex marriage. When people engage in critical thinking, they are more likely to consider diverse perspectives, weigh the pros and cons of different arguments, and make decisions based on evidence and logic rather than on biases or stereotypes.Therefore, it can be hypothesized that *people with higher levels of education are overall more in favour of same-sex marriage.*

It can also be theorized that progressive political parties tend to be more supportive of same-sex marriage and other social justice issues, while conservative parties tend to take a more traditionalist stance. Individuals who identify with progressive parties may therefore be more likely to be exposed to and adopt progressive values, including support for same-sex marriage, as they pursue higher levels of education. Conversely, individuals who identify with conservative parties may be more resistant to adopting progressive values, even with higher levels of education. *Those who value leftist political views are more likely to support same sex marriage.*

Expanding on the previous theory, gender moderates the relationship between education and support for same-sex marriage, such that the relationship may be stronger among women as compared to men. This is because women, as a historically marginalized group, may be more likely to support social justice issues, such as same-sex marriage, that challenge traditional power structures and promote equality. *Women are more likely to be in favor of same sex marriage.*

With regards to the third independent variable being studied, individuals who have higher levels of trust in others are more likely to support social justice issues, including same-sex marriage. This may be because individuals with higher levels of trust are more accepting of diverse perspectives and are more likely to support equality and human rights. Moreover, individuals with higher levels of trust may be more open to new information and ideas, which may increase their exposure to and adoption of progressive values, including support for same-sex marriage. *Those who trust in others are more likely to support same sex marriage*

To test this hypothesis, I used data collected from the 2015 Canadian Election Study, where the outcome of interest is to see if those who are **more** educated are generally more in favour of same-sex marriage.

Measurement:

In order to measure citizens' views on same-sex marriage, the dependent variable – stance towards same-sex marriage (p_ssm) – was used. To make the concept binary, p_ssm was recoded to “newssmarriage”. People were given the opportunity to choose whether they supported same-sex marriage (0) or opposed it (1).

Simultaneously, the independent variable (education) was re-coded to “neweducation” to measure the level of education someone has received. Education can profoundly influence a person's beliefs and values. A scale of 1-6 is used for this variable, which is ordinal, with 1 being a high school diploma or less, and 6 being a professional or doctorate degree.

In order to be able to accurately observe the relationship between the level of education and opinions on same-sex marriage, we had to control for some confounding variables. First, we controlled for people's religion. Religion is something that impacts both the independent variable and the dependent variable, as religious views may both dictate a person’s view on same-sex marriage, as well as the fact that many religious families may opt to send their kids into educational programs more in line with their beliefs. The next variable to control for is province, as the province in which a respondent resides may impact both their views on same-sex marriage. Certain provinces have more conservative viewpoints as opposed to others and also impact education as certain provinces may not have readily accessible forms of tertiary education.

To control for religion, we recoded the categorical variable into a binary "newreligion" based on whether one was religious (1) or not (0). The new variable simplifies the options for participants, making it easier to analyze the impact of religious beliefs on their views on same-sex marriage (CES 2015).

Lastly, the confounding concept of the province of residence of the respondent can be measured by the categorical variable (province), which we have re-coded to (newprovince). We re-coded the provinces by their geographical area, with 1 being the Maritimes, 2 being Ontario and Quebec or otherwise “Central Canada", 3 being the Prairies, 4 being British Columbia, and 5 encompassing the Territories. This study’s intent is to determine if moving westward has a positive correlation with an increasing favouring of same-sex marriage. The provinces were re-coded from 1 being the most eastward province to 5 being the farthest west.

After considering the previous variables used in the study, it was determined that three more independent variables were necessary to analyze in order to get a more holistic understanding of the study.

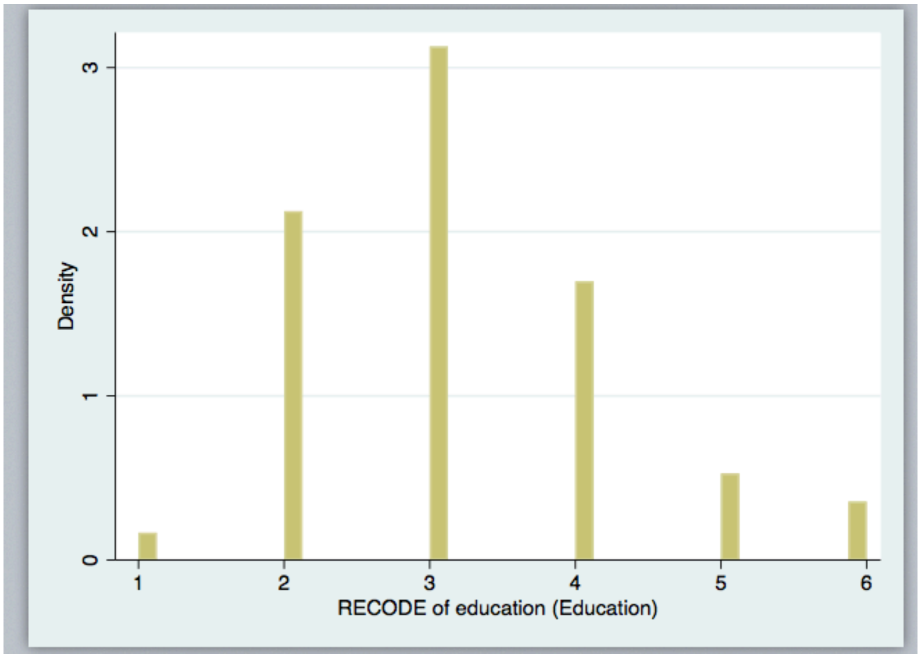
Firstly, gender might be the most relevant to consider, because it is a fundamental aspect of social identity and can influence attitudes and beliefs towards various issues, including same-sex marriage. Gender is a binary variable where 0 represents males and 1 represents females, allowing for straightforward analysis of the relationship between gender and attitudes towards same-sex marriage.

Examining the relationship between ‘vote_lean’ and attitudes towards same-sex marriage can provide valuable insights into the role of political ideology in shaping opinions on this issue. The variable vote_lean was re-coded to make a scale from most left leaning to right leaning using the 5 main parties (Bloc Quebecois, NDP, Green, Liberal Conservative) with the other options being rid of. Where 1 is the farthest left (Bloc) and 5 is farthest right (Conservative). I will look closely at the effect of one's political stance on their position on same sex marriage as political stance is an important point to consider because political ideology often shapes individuals' beliefs and values, and previous research has shown that conservative political ideology is generally associated with less support for same-sex marriage. This variable will be looked at in a crosstab later on to observe trends within the relationship.

Lastly, Research suggests that people who hold more positive attitudes towards marginalized groups, including the LGBTQ+ community, tend to have higher levels of trust in others (e.g., Pettigrew & Tropp, 2006). Respondents under the variable p_trust were surveyed based on the statement "most people can be trusted" vs. "need to be very careful when dealing with people". The variable was re-coded where 0 indicated that "most people can be trusted", and 1 indicates "you need to be very careful"

Descriptive Statistics:

From a sample of 11,064 respondents, the observed mean for "neweducation" was 3.17, with a standard deviation of 1.09, and using the standard deviation we calculated the standard error of the estimate of the mean to be 0.033, resulting in a margin of error of 0.066, meaning that we can be 95% confident that the true population mean for "neweducation" falls between 3.10 and 3.24.



This histogram shows that the variable "neweducation" follows a standard distribution with most observances near the mean, with the mean being 3.17.

A new dependent variable was created by recoding p_ssm into a binary variable (newssmarriage), representing individuals' opinions on same-sex marriage. This variable captures people's opinions on same-sex marriages. The ordinal variable in question has 5,149 observations. Individuals can either support (0) same-sex marriage or reject it (1). This variable's standard deviation is 0.43 and its mean is 0.25. The standard error of the proportion is equal to 0.007, meaning we can be 95% confident that the true population proportion of those who oppose same-sex marriage falls between 0.241 and 0.269.

The variable "newwreligion" is a binary variable representing individuals' religious affiliation, where 0 indicates that the respondent is not religious and 1 indicates that the respondent is religious. According to the new table of frequencies, out of 9,533 observations, 30.33% of respondents reported that they are not religious, while 69.67% of respondents reported that they are religious. The number of respondents who reported being not religious was divided by the total number of respondents, the standard error of the proportion is equal to 0.007, meaning we can be 95% confident that the true population proportion of those who are not religious falls between 0.288 and 0.318.

The new variable "newprovince" provides a useful demographic overview of the dataset's geographical distribution. It is important to note that the largest proportion of respondents come from Central Canada (Ontario and Quebec), accounting for approximately 64.88% of the total responses. The Prairies and British Columbia (BC) make up 13.79% and 11.42% of respondents, respectively. The Maritimes represent almost 9.81% of the respondents, while the Territories account for just 0.09% of respondents. The variable "newprovince" has a mean of 2.27 and a standard deviation of 0.79, which results in a standard error of 0.007; multiplying this value by 2 gives us a margin of error of 0.014, indicating that we can be 95% confident that the true population mean for this variable falls between 2.26 and 2.28.

Gender did not need to be recoded as the only two options were male and female. The variable "sex" indicates the respondent's gender. The dataset consists of 11,548 observations, with 5,478 (47.44%) being male and 6,070 (52.56%) being female.

Descriptive statistics for the re-coded variable p_trust (Most people can be trusted (0) vs. need to be very careful when dealing with people (1)) reveal that out of the 6,940 responses recorded, 4,101 (59.09%) individuals expressed the need to be very careful, while 2,839 (40.91%) individuals believed that most people can be trusted. A significant portion of the sample population may have a cautious attitude towards other people, which could have implications for social interactions and trust-building in the real world.

The ordinal variable "newvotelean" measures individuals' political leanings on a scale from most left-leaning to right-leaning, with 1 representing the farthest left (Bloc Quebecois) and 5 being the farthest right (Conservative). Out of the 1,092 responses after re-coding vote_lean, the mean is 3.3, indicating that the average respondent falls in the middle range between leaning towards the Green party (3) and the Liberal party (4). The standard deviation of 1.28 suggests that there is considerable variability in political leanings among the sample. Additionally, the distribution of responses shows that a little over one-third of respondents lean towards the NDP (33.79%) and the Liberal party (33.88%), while only a small percentage lean towards the Bloc Quebecois (5.68%) or the Green party (5.77%).

Correlational Analysis

```
. tab newssmarriage newvotelean
```

RECODE of p_ssm (Do you favour or oppose same-sex marriage, or do you have no op	RECODE of vote_lean (Party leaning towards) 1	2	3	4	5	Total
0	23	159	27	140	69	418
1	4	27	6	23	33	93
Total	27	186	33	163	102	511

This is a crosstab of the variables *newvotelean* and *newssmarriage.* We can observe that the most Conservative supporters have the highest proportion of opposition to same-sex marriage at 32.4%, followed by Liberal supporters at 21.5%. In contrast, the lowest proportion of opposition to same-sex marriage is among the most left (Bloc Quebecois) supporters at 9.5%. Moreover, 35.5% of those who oppose same-sex marriage belong to Conservative supporters, while 21.5% belong to Liberal supporters.

The bivariate regression (Figure A) model examines the relationship between the level of education and stance on same-sex marriage. The independent variable is the level of education on a scale of 1-6, while the dependent variable is stance on same-sex marriage (0 for those who support it and 1 for those who oppose it). The coefficient for the newssmarriage variable is -0.0400, and it is statistically significant with a p-value less than 0.05. This coefficient indicates that, on average, for each increase in level of education, the stance on same-sex marriage decreases by 4% (0 for support and 1 for opposition). Therefore, higher levels of education are associated with a more favorable stance towards same-sex marriage.

The R-squared value is 0.0101, which indicates that only a small proportion (1.01%) of the variance in stance on same-sex marriage can be explained by the level of education. However, this does not necessarily mean that the model is not useful or that the independent variable is not important. Other variables that are not included in the model may also be important in explaining the variance in the dependent variable.

B) After running a multiple regression, we can see that for every category of advancement in one's education (e.g. elementary school → high school, etc.), their opinion of same-sex marriage changes by -0.0326 (0 is in favor, 1 is opposed), meaning that the predicted probability of having a favorable opinion towards same-sex marriage increases by 3.26% for each increase in education level. This suggests that as education level increases, the opinion towards same-sex marriage tends to become more favorable. The p-value of 0.000 indicates that it is highly unlikely that the observed association occurred by chance and suggests that the relationship between education and stance on same-sex marriage is meaningful. Therefore, we can reject the null hypothesis and confidently conclude that there is a significant positive association between education level and support for same-sex marriage, and that education is a predictor of attitudes towards same-sex marriage. In order to test the relationship between education and stance on same sex marriage, I recoded education into two categories, where 0 is less than a college education and 1 is a college education or greater.

```
. ttest newweducation, by(newssmarriage)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
       0 |   3,809    .7844579    .0066635    .4112519    .7713935    .7975222
       1 |   1,268    .6774448    .0131326    .4676385    .6516808    .7032088
---------+--------------------------------------------------------------------
Combined |   5,077    .7577309    .0060138    .4284985    .7459414    .7695205
---------+--------------------------------------------------------------------
    diff |            .1070131    .0138127                .0799343    .1340918
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =   7.7475
H0: diff = 0                                     Degrees of freedom =     5075

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 0.0000
```

The t-test shows a statistically significant difference in the mean of the variable *newweducation* between those who support same-sex marriage and those who oppose it ($t = 7.7475$, $p < 0.0001$). Those who have a college education or greater (mean = 0.78) are more likely to support same-sex marriage compared to those with less than a college education (mean = 0.68). The difference in means is 0.107, with a 95% confidence interval ranging from 0.079 to 0.134. This suggests that education level is associated with attitudes towards same-sex marriage.

As we study the change in people's stance on same-sex marriage moving westward across the provinces, we can observe that one's province has a correlation to their opinions on same-sex marriage. For instance, moving west from the Maritimes to Ontario and Quebec is associated with a 3.18% increase in support for same-sex marriage; however, this relationship is not statistically significant ($p = 0.124$).

Further moving westward from Ontario and Quebec to the Prairies, we had to calculate the change in support for same-sex marriage between these regions. To do this, the coefficient of Ontario and Quebec (0.0318) was subtracted from the coefficient of the Prairies (0.0971): 0.0971 - 0.0318 = 0.0653. Thus, there is a 6.53% increase in support for same-sex marriage when moving from Ontario and Quebec to the Prairies, and this increase is statistically significant considering the significance of the Prairies coefficient ($p = 0.000$).

Continuing westward from the Prairies to BC, we need to follow the same approach by subtracting the coefficient of the Prairies (0.0971) from the coefficient of BC (0.0569): 0.0569 - 0.0971 = -0.0402. In this case, there is a 4.02% decrease in support for same-sex marriage when moving from the Prairies to BC. However, this decrease is not necessarily an accurate representation of the relationship between these two regions, as the coefficient for BC is statistically significant ($p = 0.033$), while the comparison was not directly provided in the original analysis.

Overall, the relationship between the westward movement across provinces and support for same-sex marriage appears to be somewhat complex. While there is an increase in support for same-sex marriage when moving from the Maritimes to Ontario and Quebec and then to the Prairies, there is a decrease in support when moving from the Prairies to BC. Although some of these relationships are statistically significant, it is essential to consider other variables not included in this analysis that may affect the observed relationship. Further research and a more comprehensive analysis are required to better understand the overall relationship between geographic location and opinions on same-sex marriage.

The next variable, religion, shows a one-unit change in religion (0 is not religious, 1 is religious) correlates to a 0.2218 increase towards negative opinions on same-sex marriage. With a high t-stat of 16.18, we can also safely conclude that the effect of religion on opinions towards same-sex marriage is statistically significant at the 95% confidence level. However, more research is required to be able to better interpret this data

Based on figure 3, we see that after including three additional independent variables in the regression analysis, we can observe the following results. Firstly, the coefficient for level of education (neweducation) is no longer statistically significant ($p = 0.717$). This suggests that, when controlling for other variables, education level is no longer a significant predictor of attitudes towards same-sex marriage. However, the coefficients for newwreligion ($p = 0.000$), newwsex ($p = 0.005$), and newvotelean ($p = 0.029$) are all statistically significant.

The coefficient for newwreligion indicates that being religious is associated with a 15.44% decrease in support for same-sex marriage. The coefficient for newwsex suggests that being female is associated with a 10.19% decrease in support for same-sex marriage. Lastly, the coefficient for newvotelean indicates that being more right-leaning in one's political views is associated with a 3.13% increase in support for same-sex marriage. Lastly, the multiple regression results show that the coefficient for "newtrust" is positive (0.0503), indicating that higher levels of trust in others are associated with a more favorable stance towards same-sex marriage. However, this relationship is not statistically significant ($p = 0.165$).

Summary

The previous data analysis sought to examine the relationship between citizens' levels of education and their attitudes towards same-sex marriage, using data from the 2015 Canadian Election Study. The initial results indicated that there is a significant positive association between education level and support for same-sex marriage. As respondents' education level increases, they are more likely to have a favorable opinion of same-sex marriage. However, the study also found that this relationship is complex and may be influenced by other factors, such as geographic location and religious beliefs. Moving westward across provinces showed an increase in support for same-sex marriage from the Maritimes to the Prairies, but a decrease in support from the Prairies to British Columbia. Additionally, religious individuals were more likely to have negative opinions on same-sex marriage.

The re-constructed analysis examined the relationship between education level, trust, religion, gender, and political leaning with attitudes towards same-sex marriage in Canada. The crosstab and regression models demonstrate the relationship between independent variables and opinions on same-sex marriage. The results indicate that education level is no longer a significant predictor of attitudes towards same-sex marriage when controlling for other variables. The study also found that being female was associated with a decrease in support for same-sex marriage, while being more right-leaning in one's political views was associated with an increase in support for same-sex marriage. Additionally, higher levels of trust in others were associated with a more favorable stance towards same-sex marriage, although this relationship was not statistically significant. Overall, the analysis provides a comprehensive overview of the relationship between different demographic, social, and political factors and opinions on same-sex marriage in Canada.

In conclusion, although the relationship between education and attitudes towards same-sex marriage is apparent, the analysis indicates that further research and a more comprehensive examination of other factors is necessary to fully understand the nuances of this relationship.

Works Cited

Canadian Election Study 2015 Web and Phone Combined File Codebook and Questionnaire Contents.

Appendix

Figure 1 Bivariate regression

```
. regress newssmarriage neweducation

      Source |       SS           df       MS      Number of obs   =     5,077
-------------+----------------------------------   F(1, 5075)      =     51.65
       Model |  9.58512512         1  9.58512512   Prob > F        =    0.0000
    Residual |  941.727067     5,075  .185561984   R-squared       =    0.0101
-------------+----------------------------------   Adj R-squared   =    0.0099
       Total |  951.312192     5,076  .187413749   Root MSE        =    .43077

------------------------------------------------------------------------------
newssmarri~e | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
neweducation |  -.0399971   .0055651    -7.19   0.000    -.0509071   -.0290871
       _cons |   .3810896   .0192479    19.80   0.000     .3433554    .4188237
------------------------------------------------------------------------------
```

Figure 2 Updated Multiple regression from A3

```
. regress newssmarriage neweducation i.newprovince newwreligion

      Source |       SS           df       MS      Number of obs   =     4,386
-------------+----------------------------------   F(5, 4380)      =     63.70
       Model |  55.7608549         5  11.152171   Prob > F        =    0.0000
    Residual |  766.864088     4,380  .175083125   R-squared       =    0.0678
-------------+----------------------------------   Adj R-squared   =    0.0667
       Total |  822.624943     4,385  .187599759   Root MSE        =    .41843

------------------------------------------------------------------------------
newssmarri~e | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
neweducation |  -.0326174   .0058475    -5.58   0.000    -.0440815   -.0211532
             |
 newprovince |
          2  |   .0317975   .0206655     1.54   0.124    -.0087173    .0723124
          3  |   .0970681   .0255236     3.80   0.000      .047029    .1471072
          4  |   .0568726   .0266874     2.13   0.033     .0045518    .1091935
             |
newwreligion |   .2217745   .0137035    16.18   0.000     .1949086    .2486403
       _cons |   .1658547   .0292311     5.67   0.000     .1085468    .2231625
------------------------------------------------------------------------------
```

Figure 3: Regress all variables

```
. regress newssmarriage neweducation i.newprovince newwreligion newtrust newwsex newvotelean

      Source |       SS           df       MS      Number of obs   =       423
-------------+----------------------------------   F(8, 414)       =      4.29
       Model |  4.57135161         8  .571418951   Prob > F        =    0.0001
    Residual |  55.1733292       414  .133268911   R-squared       =    0.0765
-------------+----------------------------------   Adj R-squared   =    0.0587
       Total |  59.7446809       422  .141575073   Root MSE        =    .36506

------------------------------------------------------------------------------
newssmarri~e | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
neweducation |   .0063837   .0175938     0.36   0.717    -.0282005    .0409679
             |
 newprovince |
          2  |  -.0058002   .0550824    -0.11   0.916    -.1140763    .1024759
          3  |    .068214   .0729941     0.93   0.351    -.0752712    .2116993
          4  |  -.0201306   .0727799    -0.28   0.782    -.1631948    .1229335
             |
newwreligion |   .1544071   .0385659     4.00   0.000     .0785977    .2302165
    newtrust |   .0502976    .036171     1.39   0.165    -.0208042    .1213994
     newwsex |  -.1019486   .0364896    -2.79   0.005    -.1736766   -.0302206
 newvotelean |   .0313419   .0143306     2.19   0.029     .0031721    .0595117
       _cons |  -.0215521   .1022278    -0.21   0.833    -.2225023    .1793982
------------------------------------------------------------------------------
```